

WILDFIRE RISK ASSESSMENT REPORT

Machine-Learning-Based Environmental Risk Assessment

1. Assessment Information

Prediction ID:	PRED-1234567890
Assessment Date:	2026-10-10
Generated Date:	10 October 2026, 08:29 UTC
Assessment Location:	Sinharaja Forest Reserve, Sri Lanka
Coordinates:	6.42182, 80.487502
Model:	XGBoost

2. Prediction Result



3. Environmental Conditions

3.1 Location

Location Name:	Sinharaja Forest Reserve, Sri Lanka
Latitude:	6.42182
Longitude:	80.487502

3.2 Weather Conditions

Precipitation:	12.5 mm
Minimum Relative Humidity:	45 %
Maximum Relative Humidity:	85 %
Specific Humidity:	0.015
Solar Radiation:	150.0 W/m²
Minimum Temperature:	22.5 °C
Maximum Temperature:	32.1 °C
Wind Speed:	4.2 m/s

3.3 Fire / Fuel Conditions

Burning Index:	45.2
Fuel Moisture 100hr:	12 %
Fuel Moisture 1000hr:	14 %
Energy Release Component:	33

3.4 Atmospheric Conditions

Reference Evapotranspiration:	4.2
Potential Evapotranspiration:	5.1
Vapor Pressure Deficit:	1.2 kPa

3.5 Derived Date Features

Year:	2026
Month:	10
Day of Year:	283

4. Risk Interpretation

Moderate wildfire risk detected. Increased awareness and monitoring are recommended.

5. Recommended Actions

- ✓ Increase monitoring.
- ✓ Avoid unnecessary outdoor burning.
- ✓ Review preparedness.
- ✓ Monitor official warnings.

***Disclaimer:** This prediction is a machine-learning estimate based on the supplied environmental and geographical conditions. It should not replace official emergency warnings, professional wildfire assessments, or emergency-management instructions.*